

Bill Zhang

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EDUCATION

University of Southern California

Master of Science in Computer Science - Artificial Intelligence

August 2023 – May 2025

Los Angeles, CA

University of California, Santa Cruz

Bachelor of Science in Computer Science

September 2020 – March 2023

Santa Cruz, CA

EXPERIENCE

Applied AI Engineer (Forward-Deployed)

Scale AI

June 2025 – July 2026

San Francisco, CA

- Shipped a production **multi-agent LLM system automating denied-claim investigations** for a PE-backed healthcare-claims company: an orchestrator routing **3 specialized sub-agents** across **20 denial-code policy workflows**, serving **300–500 weekly users** on up to **20K claims/week**.
- Designed and owned the **LLM-as-judge evaluation framework** (3-dimension rubric + error-analysis loop) that raised QA-audit accuracy **64% → 88%**; quality results helped close a **seven-figure** contract expansion.
- Owned evaluation, quality, and cost for an **LLM clinical decision-support agent at Mayo Clinic** (eight-figure account); built per-patient token/cost accounting and cut full validation-run cost $\sim 4\times$ (\$4.6K $\rightarrow$ $\sim$ \$1.2K), validating a **23 $\times$ -cheaper** critic model via A/B evals.
- Root-caused a scoring-rubric artifact that was masking true agent quality at Mayo, recovering the evidence-grounding metric **81 → 93**; rebuilt the clinical citation pipeline from **$\sim 0\%$ resolved sources to fully resolved**, clearing a pilot blocker.
- Led agent design for the **TIME AI Agent** (publicly launched Nov 2025); built the agent loop, tool use, and evals for RAG search, summarization, and **13-language** translation across TIME's 102-year archive, with production guardrails raising adversarial prompt blocking **30% → 95%**.
- Hardened Cengage's production EdTech agents against **prompt-injection and jailbreak attacks**, closing **every finding** from an internal red-team audit and supporting a **multi-million-dollar** account expansion.

AI Engineer

RingCentral

August 2023 – June 2025

Belmont, CA

- Built an LLM analytics + eval pipeline (OpenAI, Pandas, GCP) that expanded coverage of **10,000+ weekly** support chats from **1–2% → 100%**; cut processing time **8 hours → 1 hour**.
- Improved accuracy **90% → 95%** and cut agent hand-offs **40% → 13%** by adding rubric-based retrieval scoring and weekly monitoring (**20 Looker** dashboards).

PROJECTS

Slug Loop | *React, Material UI, ExpressJS, C, Azure*

March 2023 – Present

- Served **20,000+ users** by shipping a real-time UCSC bus tracking web app with a React frontend, ExpressJS APIs, and Azure deployment.
- Placed **Global Top 10** in the Google Solution Challenge out of **2,000+ projects** (2023).

DispatchAI | *NextJS, FastAPI, Twilio, RetellAI, Hume, Google Maps*

June 2024 – Present

- Built and demoed a low-latency voice agent for emergency call intake + escalation (Twilio + FastAPI + NextJS), including intent routing and structured handoff to responders.
- Won **Grand Prize (\$50k)** at UC Berkeley AI Hackathon (**900+ participants**) by integrating RetellAI + Hume and deploying a real-time pipeline (fine-tuned Mistral 7B).

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, SQL, Bash

LLM / Agents: multi-agent orchestration, LLM-as-judge & evaluation design, RAG, prompt engineering, prompt-injection/jailbreak defense, cost & latency optimization

Models: OpenAI GPT, Google Gemini / Vertex AI, Anthropic Claude

Infra/Web: GCP, Azure/AKS, Kubernetes, Docker, CI/CD, FastAPI, React, NextJS, PostgreSQL